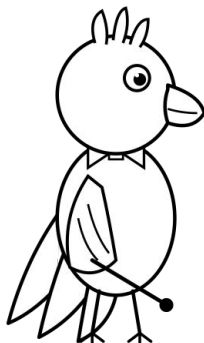


Grade 2

Ontario Curriculum

Algebra (Patterns and Relationships)

The Pattern Parade — Growing, Shrinking, and
Numbers to 100



Unit Overview: The Pattern Parade — Growing, Shrinking, and Numbers to 100

Grade: Grade 2 | **Strand:** Algebra (Patterns and Relationships)

Target Expectations: C1.1, C1.2, C1.3, C1.4

Learning Goal: Children will identify, describe, and create repeating, growing, and shrinking patterns; translate them across shapes, numbers, and tables of values; and use pattern rules to predict, justify, and find missing terms with whole numbers up to 100.

Lesson	Lesson Title	Learning Goal
Day 1	Welcome Back to the Parade!	I can identify a pattern as repeating, growing, or shrinking and describe what changes from one term to the next.
Day 2	Growing and Shrinking Parades	I can build a growing parade and a shrinking parade and tell my partner the rule for each.
Day 3	The Parade Table of Values	I can take one pattern and show it as a parade, as numbers, AND as a table of values.
Day 4	Pattern Detective Pro	I can find a missing term in a growing or shrinking pattern and use the rule to predict a faraway term.
Day 5	Lead the Number Parade to 100	I can build a number parade up to 100, name its rule, organize it as a table of values, and lead it.

Lesson 1: Welcome Back to the Parade!

Goal: Patterns can repeat, grow, or shrink — we can spot all three kinds in our classroom and our world.

Expectation: C1.1

Learning Goal: "I can identify a pattern as repeating, growing, or shrinking and describe what changes from one term to the next."

Materials: M1_animal_cards, M2_parade_strip, M3_sound_movement_cards, M6_anchor_chart_words

1. Minds On (12 mins): Coco's Three Mystery Parades

Coco brought THREE different parades today. They look different — and Coco says they are different KINDS of patterns. Our job is to figure out the three kinds.

Bring children to the carpet in a circle. Hold up Coco. Say to students: 'Welcome back, parade builders! Last year — or in Grade 1 next door — we learned about repeating patterns. Coco taught us rules like AB and ABB. This year, Coco has TWO new tricks. Let me show you.'

2. Action (28 mins): The Three-Pattern Hunt

- Whole-group sorting demo: lay out three more pre-built parades (one of each type) and three small chart-paper labels: REPEATING, GROWING, SHRINKING.
- Partner build: each pair gets one set of Animal Cards (M1) and one Parade Strip (M2).
- Real-life pattern hunt (10 min): pairs walk the classroom (and corridor if safe) with a clipboard.
- Whole-class share: pairs return to the carpet and share one find each.
- Circulate during partner build and the hunt.

3. Consolidation (10 mins): Three-Type Share Circle

Discuss:

- What was the MOST interesting real-life pattern you found today?
- Did you find any pattern that was hard to label?
- If a pattern keeps growing, when does it stop?

Exit: On the way out the door, each child gives the teacher one example of each type — 'A repeating pattern is ____.

Lesson 2: Growing and Shrinking Parades

Goal: In a growing pattern, each term gets bigger by the same amount; in a shrinking pattern, each term gets smaller by the same amount.

Expectation: C1.1

Learning Goal: "I can build a growing parade and a shrinking parade and tell my partner the rule for each."

Materials: M1_animal_cards, M2_parade_strip, M5_core_loop, M6_anchor_chart_words

1. Minds On (12 mins): Coco's Rule-Naming Game

Yesterday we said 'getting bigger' and 'one more.' Today we give the rule a NAME using a + or a - and a number.

Bring children to the carpet. Lay parade 1 (growing +2: 2 Beas, 4 Beas, 6 Beas, 8 Beas across 4 cells). Hold up Coco.

2. Action (28 mins): Build the Rule, Loop the Step

- Whole group demo: build a growing +3 parade slowly with the class (3 Beas, 6 Beas, 9 Beas, 12 Beas).
- Partner build round 1: each pair gets Animal Cards (M1), one Parade Strip (M2), one Core Loop Frame (M5), and a stack of sticky notes.
- Partner build round 2: now build TWO shrinking parades — one with a -1 rule, one with a -2 rule.
- Stretch challenge for fast pairs: invent a rule the teacher has not shown yet (e.g., +5, -3, +10, +0 special case).
- Circulate.

3. Consolidation (10 mins): Rule Showcase Round

Discuss:

- Who built a growing parade with the BIGGEST jump?
- Did anyone build a shrinking parade where the rule made the parade end exactly at zero?
- Yesterday we sorted patterns into three types.

Exit: On the way out the door, each child names one growing rule and one shrinking rule using +N or -N notation ('My growing rule is +___).

Lesson 3: The Parade Table of Values

Goal: We can show the same pattern as a parade, as a sequence of numbers, or as a table of values — and the rule is the same in all three.

Expectation: C1.2

Learning Goal: "I can take one pattern and show it as a parade, as numbers, AND as a table of values."

Materials: M1_animal_cards, M2_parade_strip, M5_core_loop

1. Minds On (12 mins): Coco's Three Faces of One Pattern

Coco built ONE pattern today and put on THREE different costumes. We have to find out: are these really the same pattern?

Bring children to the carpet. Lay all three representations in a horizontal line on the floor: the animal parade (3, 6, 9, 12 Beas), the number sequence sentence strip (3, 6, 9, 12), and the chart-paper table of values.

2. Action (28 mins): Translate Across Three

- Whole-class table-building demo: take a fresh growing pattern (the class chooses the rule, e.g., +5) and build it together.
- Partner translation round 1: each pair gets Animal Cards (M1), one Parade Strip (M2), one Core Loop Frame (M5), and a piece of paper or small whiteboard.
- Partner translation round 2: now a SHRINKING rule (-1, -2, -3).
- See the expanded walkthrough below for the keystone reveal moment when pairs realize the table of values reveals the rule even when the parade and number sequence get long.
- Circulate.

3. Consolidation (10 mins): Three-Rep Showcase

Discuss:

- Who would like to show their THREE representations to the class?
- Which representation was EASIEST to build?
- Look at the table of values.

Exit: Each child whispers to a partner: 'My rule was ____, and I showed it three ways: as a parade, as numbers, and as a table.' Then tidy.

Lesson 4: Pattern Detective Pro

Goal: Once we know the rule, we can find missing terms anywhere AND predict terms far ahead — even term 20 — without drawing every step.

Expectation: C1.3

Learning Goal: "I can find a missing term in a growing or shrinking pattern and use the rule to predict a faraway term."

Materials: M1_animal_cards, M2_parade_strip, M4_missing_card, M5_core_loop

1. Minds On (12 mins): Coco's Mystery Cases

Coco is a detective today, and he has THREE cases to crack. One repeating, one growing, one shrinking — and each one is missing a term. Can we use the rule to find the missing piece AND predict what comes way ahead?

Bring children to the carpet. Lay the three broken patterns in a line. Hold up Coco.

2. Action (28 mins): Detectives at Work — Find and Predict

- Whole-group far-prediction demo: take a fresh growing +4 pattern (4 Beas, 8 Beas, 12 Beas across 3 cells).
- Partner round 1 — find missing across types: each pair gets Animal Cards (M1), one Parade Strip (M2), 5 Missing Mystery Cards (M4), and one Core Loop Frame (M5).
- Partner round 2 — far prediction: each pair builds a fresh growing or shrinking pattern across 4 terms.
- Stretch challenge for fast pairs: trade pattern starts with another pair.
- Circulate.

3. Consolidation (10 mins): Detective Pro Share Circle

Discuss:

- Who solved a tricky pattern today?
- Was it harder to find the missing term in a REPEATING pattern, a GROWING pattern, or a SHRINKING pattern?
- If your growing pattern had a +5 rule starting at 5, what would term 100 be?

Exit: On the way out, each child says one full justify sentence using rule language: 'I am a Pattern Detective Pro because I can use the +___ or -___ rule to predict term ___ without drawing it.' Teacher...

Lesson 5: Lead the Number Parade to 100

Goal: Pattern rules also work with bigger numbers — we can build repeating, growing, and shrinking number parades up to 100.

Expectation: C1.4

Learning Goal: "I can build a number parade up to 100, name its rule, organize it as a table of values, and lead it."

Materials: M1_animal_cards, M2_parade_strip, M3_sound_movement_cards, M4_missing_card

1. Minds On (12 mins): From Animals to Numbers up to 100

All week we've built parades. Today Coco asks the BIG question: can the rule work with numbers all the way up to 100? And can YOU lead a parade?

Bring children to the carpet. Hold up Coco. Beside him, lay the small +3 example (3, 6, 9, 12) AND the bigger +10 example (10, 20, 30, 40) parallel.

2. Action (40 mins): Plan, Build, Translate, Table, Lead

- Phase 1 — PLAN (5 min).
- Phase 2 — BUILD an animal parade (8 min).
- Phase 3 — TRANSLATE to numbers up to 100 (8 min).
- Phase 4 — TABLE (7 min).
- Phase 5 — LEAD (5 min).
- Phase 6 — REVISE & REFLECT (8 min).

3. Consolidation (10 mins): Junior Pattern Conductor Ceremony

Discuss:

- Who would like to LEAD the WHOLE class through their parade? (Take 2-3 children.
- What was the trickiest part of making a NUMBER parade up to 100 compared to the smaller parades earlier in the unit?
- What was your favourite parade you built this WEEK — across all 5 days — and what was its rule and type?

Exit: Each child receives a Junior Pattern Conductor certificate (the teacher reads it briefly at the front and hands it over with a high five).

Name: _____

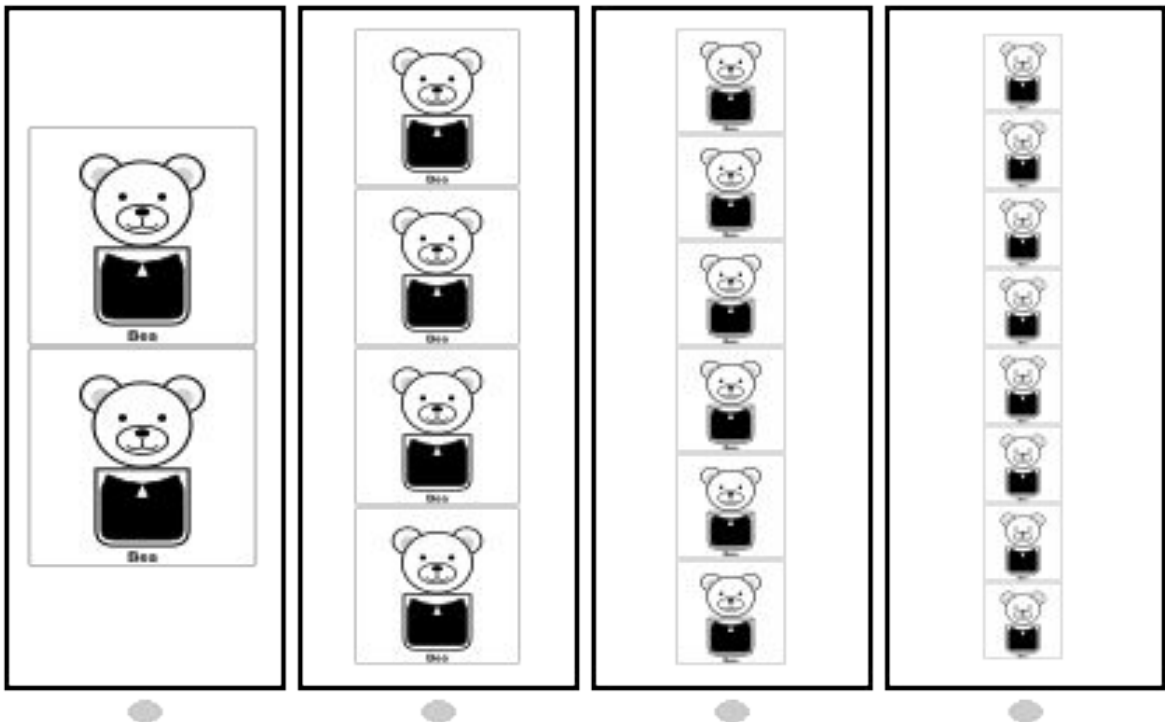
Date: _____

Coco's Three-Type Sorter

Learning Goal: I can identify a pattern as repeating, growing, or shrinking and describe what changes from one term to the next.

Instructions for the Student:

1. Look at each parade strip.
2. Look at this growing parade.
3. Look at each real-life thumbnail.



Name: _____

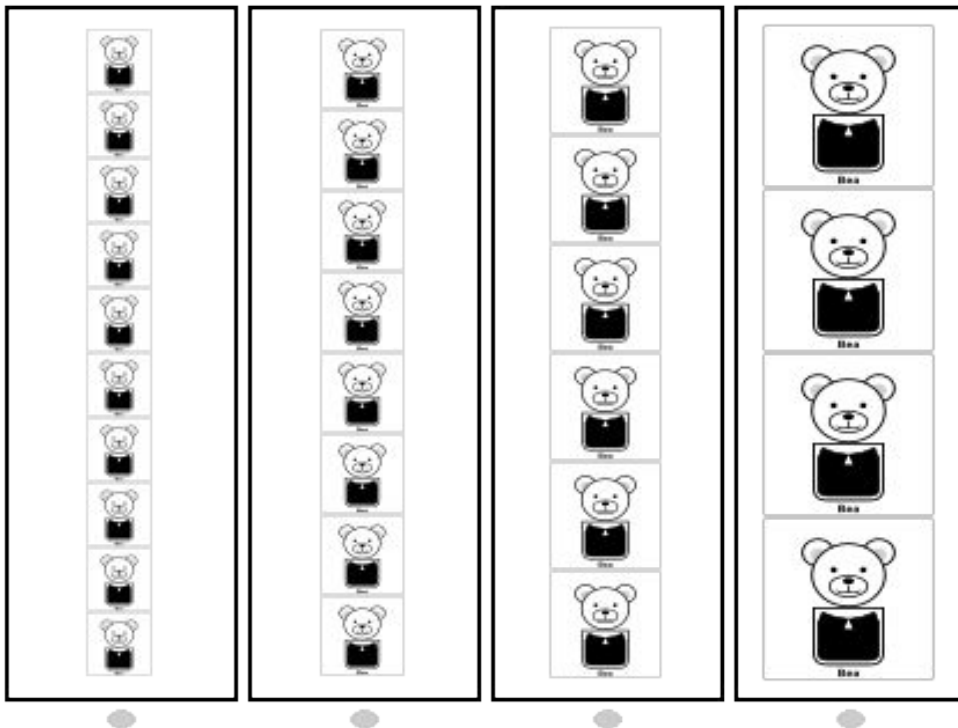
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Name the Jump

Learning Goal: I can build a growing parade and a shrinking parade and tell my partner the rule for each.

Instructions for the Student:

1. Look at each parade strip.
2. Pick a starting number.
3. Pick your own shrinking rule and starting number.



(3 parts on the printable worksheet)

Name: _____

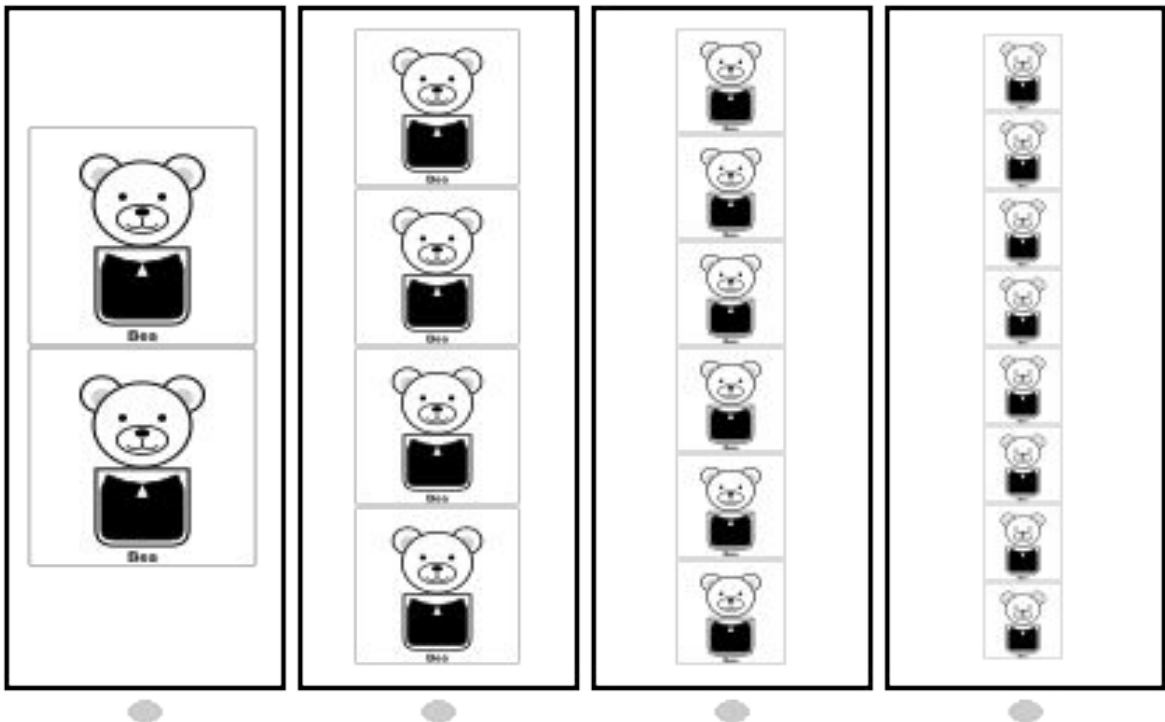
Date: _____

Three Faces of a Rule

Learning Goal: I can take one pattern and show it as a parade, as numbers, AND as a table of values.

Mission

1. Look at the growing parade at the top.
2. This table is missing one value.
3. Pick your own $+N$ rule (or $-N$ for shrinking).



(3 parts on the printable worksheet)

Name: _____

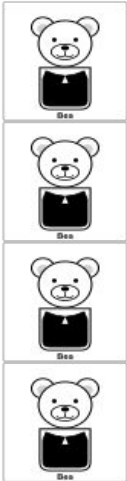
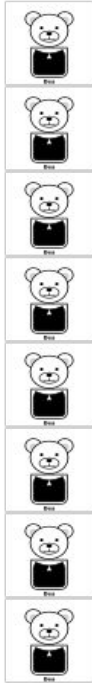
Date: _____

Detective Pro Cases

Learning Goal: I can find a missing term in a growing or shrinking pattern and use the rule to predict a faraway term.

Instructions for the Student:

1. Each pattern is missing one term.
2. Use the table to find the missing values AND predict term 10 using the rule.
3. Coco's parade has rule $+6$ starting at 6.

			
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(3 parts on the printable worksheet)

Name: _____

Date: _____

Lead the Number Parade to 100

Learning Goal: I can build a number parade up to 100, name its rule, organize it as a table of values, and lead it.

Instructions for the Student:

1. You are the Conductor today.
2. Use your rule to fill all 8 cells of the animal parade.
3. Fill in the Value column of the table with your 8 number-parade values.

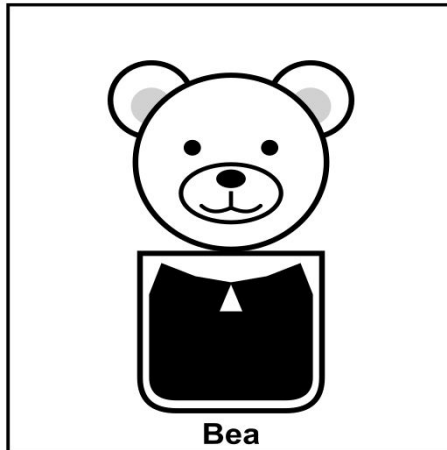
0–100 chart

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99
100									

Parade Animal Cards

(M1_animal_cards)

Reusable physical cards used by pairs to build, manipulate, read, translate, and fix parade patterns.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 2 sets (32 cards) per pair

Lessons: 1, 2, 3, 4, 5

Prep: ~30 min

TEACHER PREP

1. Print 2 sets per pair for Grade 2 (one set is 16 cards = ~10 sets total for a.

2. Optional but strongly recommended.

3. Cut along the thin solid lines into 16 individual 3in x 3in cards per set.

4. Store each pair's 32-card set in a labelled zip-top bag or envelope ('Parade.

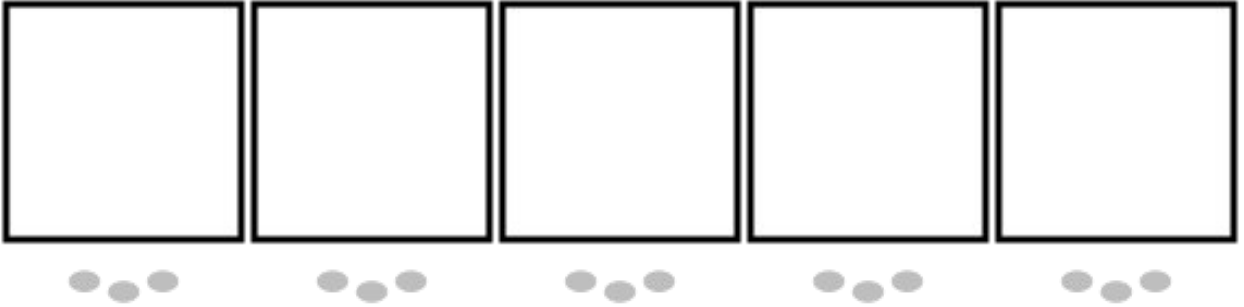
5. Grade 2 storage tip: separate bags by animal (one bag of all Beas, one of all Finns,...

...

Blank Parade Strip

(M2_parade_strip)

A printable strip with 5 cells.



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 strip per pair plus 2 spare for the teacher's whole-g

Lessons: 1, 2, 3, 4, 5

Prep: ~15 min

TEACHER PREP

1. Print enough pages to yield ~12 strips (one per pair plus 2 spare = 6 pages).

2. Laminate the pages.

3. Cut along the horizontal cut line between the two strips on each page.

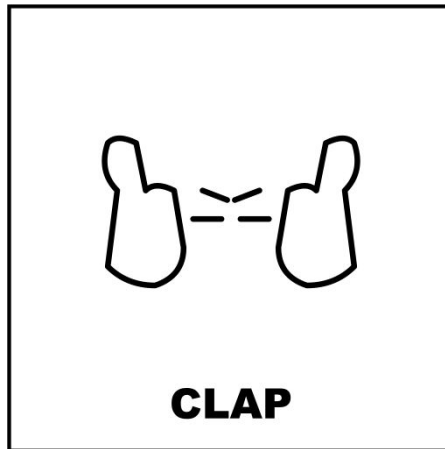
4. Distribute one strip per pair.

5. Bring a class set of dry-erase markers and small cloths/erasers.

Sound and Movement Cards

(M3_sound_movement_cards)

An 8-card set used by the class and by pairs to translate visual patterns into auditory and kinesthetic patterns.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 teacher set (8 cards) + 1 spare

Lessons: 1, 5

Prep: ~10 min

TEACHER PREP

1. Print 2 pages.

2. Laminate the pages for durability.

3. Cut along the cut-lines into 8 individual cards per set.

4. Store each set in a labelled envelope.

5. Optional colour suggestion: B&W works well; print in colour for extra visual variety.

Missing Mystery Card

(M4_missing_card)

A small card showing a question mark inside a dotted outline.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 5 cards per pair

Lessons: 4, 5

Prep: ~15 min

TEACHER PREP

1. Print 5 pages (60 cards = 5 cards per pair × 10 pairs + spares).

2. Laminate before cutting.

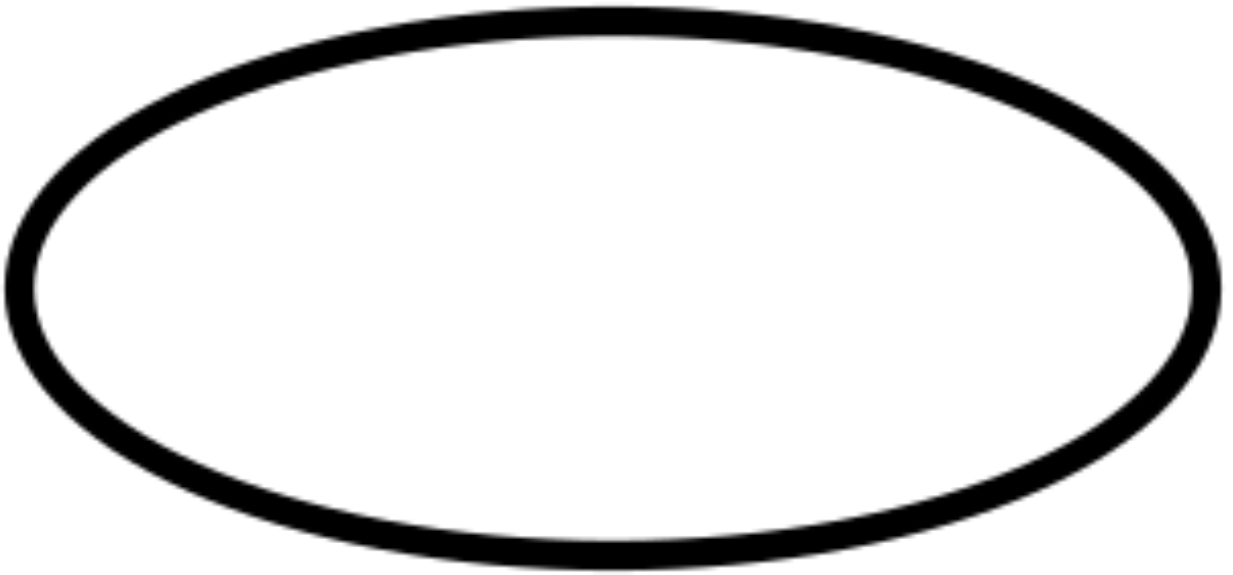
3. Cut along the cut-lines into individual 2in × 2in cards.

4. Distribute 5 cards per pair before Lesson 4.

Core Loop Frame

(M5_core_loop)

A reusable physical frame.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 frame per pair plus 2 spares

Lessons: 2, 3, 4, 5

Prep: ~15 min

TEACHER PREP

1. Print on cardstock if available.

2. Alternative: print on regular paper and laminate.

3. Cut around the outside of each oval to produce 4 frames per page (3 pages = 12).

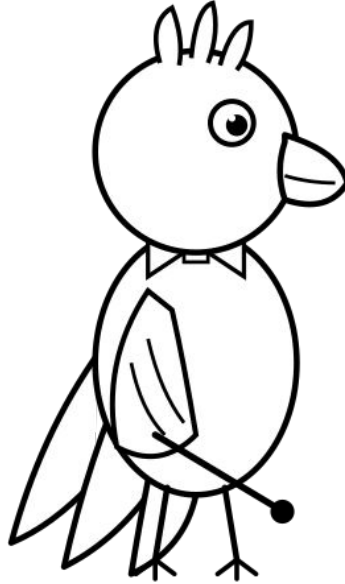
4. Distribute one frame per pair before Lesson 2.

5. Provide a small pad of sticky notes per pair.

Coco the Conductor Puppet

(char_coco_puppet)

A printable Coco illustration that the teacher mounts on a popsicle stick to use as the recurring class character throughout all 5 lessons.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet for the teacher

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock if possible (or regular paper + laminate for durability).

2. Cut around the outside of Coco's silhouette including the bottom tab.

3. Glue or tape a popsicle stick or tongue depressor behind the bottom tab so Coco.

4. Optional: laminate before cutting for years of reuse.

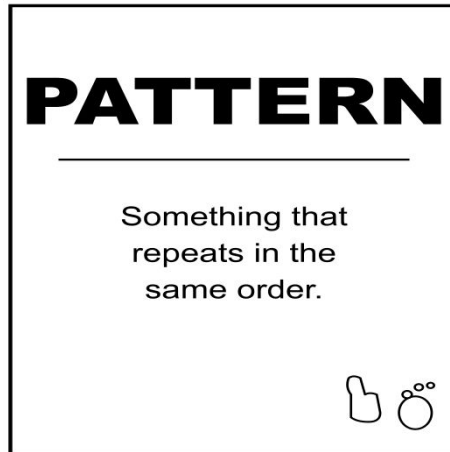
5. Store flat in a folder so the puppet doesn't crease.

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Anchor Chart Word Cards

(M6_anchor_chart_words)

A printable set of four word cards (PATTERN, CORE, MISSING, EXTEND).



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 set per class

Lessons: 1, 2, 4

Prep: ~12 min

TEACHER PREP

1. Print 1 page (1 set per class is enough — these go on the wall).
2. Laminate the page for durability before cutting.
3. Cut along the cut-lines into 4 individual word cards.
4. On Day 1, after introducing 'pattern', tape the PATTERN card to your anchor chart paper.
5. On Day 2, after introducing rule notation, tape the CORE card.

...

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C1.1 Identify repeating, growing, shrinking patterns	Identifies pattern type only with prompts.	Identifies the 3 types most of the time; informal language.	Confidently identifies repeating, growing, shrinking; spots in real life.	Identifies pattern types in unfamiliar contexts; combined types.
C1.2 Translate across reps (shapes, numbers, tables of values)	Treats reps as different patterns; no table of values.	Translates with help; table may miss headers or mismatch.	Translates across parade, numbers, and table of values.	Translates into invented reps; uses table to predict relationships.
C1.3 Determine rules (+N/-N); justify; find missing in any type	Cannot name +N / -N rule; fills missing randomly.	Names +N / -N with prompts; fills missing at end only.	Names rule, fills missing anywhere, predicts term 10 by skip-counting.	Predicts term 20+ efficiently; recognizes 100-ceiling constraints.
C1.4 Use patterns to describe whole numbers up to 100	Does not yet build a number parade up to 100.	Builds with small numbers (1-20); relationship unclear.	Builds number parade up to 100; plans starting number; explains relationship.	Uses larger rules; recognizes factors of 100; generalizes term-value relationship.

Junior Pattern Conductor Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Pattern Parade unit, identifying repeating, growing, and shrinking patterns, building tables of values, and leading a number parade up to 100. A pattern parade certificate. Coco the Conductor congratulates the student.

By completing this unit, this student has demonstrated:

- Identifying patterns as repeating, growing, or shrinking — including in real-life contexts
 - Naming pattern rules using +N or -N notation
 - Translating one pattern across animal parade, number sequence, AND table of values
- Finding missing terms anywhere AND predicting faraway terms...



Marketplace Listing

A 5-day Ontario Grade 2 unit on Algebra (C1.1–C1.4), framed as a parade led by Coco the Conductor and four Canadian animals.

Price: \$13.99 CAD

Pages: ~42

Classroom time: 255 min

Teacher prep: ~110 min

Top tags: patterns, grade 2, Ontario, Ontario curriculum, math, mathematics, algebra, patterns and relationships

What's included:

- 5 fully-scripted lesson plans (50–55 minutes each, ~255 minutes total of teaching time)
- 5 student worksheets aligned to each lesson, with name/date fields, child-voice learning goals, and Grade 2 sentence frames for justification
 - 1 mid-unit formative assessment ('The Three Pattern Types Check') administered between Lesson 3 and Lesson 4
 - 1 end-of-unit reflection sheet with 5 prompts for student self-assessment, including a yes/no audit of all four Grade 2 expectations
- 7 printable teacher manipulatives: parade animal cards (2 sets per pair to support growing patterns), parade strips (laminated for dry-erase number parades up...)
- 4 class observation trackers: 1 diagnostic (Lesson 1) plus 3 formative (Lessons 2, 3, 4) — each with grade-2-specific column headers including +N notation,...
- 1 four-level summative rubric covering all four expectations (C1.1–C1.4) with Grade 2 descriptors aligned to the 0-100 number range and table-of-values...
- 1 summative administration guide tying the worksheet, lesson, and rubric together for the teacher
 - Junior Pattern Conductor certificate (one per child) for the closing ceremony
 - Detailed expanded walkthroughs for Lesson 3 (the keystone — table of values reveal) and Lesson 5 (animals to numbers up to 100), including expected student...
 - Differentiation tracks built into every lesson: emerging learners, extending learners, ELL/IEP, and movement-first learners
 - Real-life pattern hunt (Lesson 1) with Canadian classroom contexts — calendar, hopscotch, brick walls, snowflakes